4. Analysis

4.1. Statistical Analysis

Table 1 contains the result of the Equation 1 regression with 2015 as the reference year[7]. The regression results for the remaining iterations through each reference year may be found in Table 14 in Appendix A.

Table 1. Equation 1 Regression Results[8]

Regression Variable Coefficient	Coef.	Robust Std. Err.	t	P>t	[95% Conf. Interval]	
Size ($/kW$_{DC}$)	2.943	.0673	43.74	<0.000	2.811	3.075
Year#Size						
2016 ($/kW$_{DC}$)	0.0607	0.0636	0.95	0.340	-0.0640	0.185
2017 ($/kW$_{DC}$)	-0.339	0.0653	-5.19	<0.000	-0.467	-0.211
2018 ($/kW$_{DC}$)	-0.119	0.0623	-1.90	0.057	-0.241	0.00351
Constant ($)	1409	215.0	6.55	<0.000	987.2	1830

The results suggest that 2016 marginal costs are not statistically different than 2015, while marginal costs are statistically different for 2017 and 2018.[9] Rotating the reference year produces significant differences for all marginal costs except for the 2015 and 2016 result found already (see Table 14). The trend displayed in the data is a constant marginal cost from 2015 to 2016 followed by a drop in 2017 which shows no statistically significant change in 2018. The fact that the p-value for 2018 is significant at a 10 % level suggests it is worth investigating in a future model with a larger data set, especially considering that the coefficient for 2018 is larger when compared to 2017 and could be an effect of the solar tariffs implemented in 2018. However, it fails to meet the determined cutoff (5 % level of significance) for significance here and is not discussed further.

An analysis was also run for the total average cost per watt by year for comparison purposes to the non-linear cost model (Table 2). Like the Equation 1 regression, the results in Table 2 do not account for differences in quality or county. These results show a similar trend, with 2015 and 2016 being similar, a decrease from 2016 to 2017, followed by an increase in 2018. The primary difference is that using the total average cost per watt[10] ($\hat{X}$) assumes that the entirety of the PV system price is marginal, ignoring any fixed costs of installation. Therefore, the mean costs are consistently higher (6 % to 9 %) than the marginal cost estimates using the cost function developed in Table 1. The focus of this paper is not on total average cost per watt (as reported in prior studies) and no difference in means hypothesis testing was completed. Thus Table 2 is meant only for a cursory comparison.

[7] The R^2 value is 0.9829
[8] R^2 is equal to 0.9387
[9] At a 10 % LoS
[10] As based on the statistical mean in lieu of the conditional mean calculated in the regression

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Table 2. Statistical Moments of marginal cost by year

Year	Average Size	Standard Deviation Size	Average USD/W	Stand. Dev. USD/W	Marginal Cost from Regression	Difference between Average USD/W and Marginal Cost from Regression
2015	7695	4255	3.21	0.36	2.94	0.27 (9 %)
2016	9371	4297	3.18	0.32	3.00	0.18 (6 %)
2017	9203	4317	2.75	0.40	2.60	0.15 (6 %)
2018	10 454	5238	3.02	0.36	2.82	0.20 (7 %)

The results of the regression in Equation 2 are provided in Table 3 (statistically significant coefficients are shaded).[11] For reference there are 1108 *Standard* panel systems and 1026 *Premium* panel systems. Between counties there are 73 systems in AC, 517 systems in DC, 622 systems in FC, 639 systems in MC, and 306 systems in PG.

[11] R^2 value equals 0.9614

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Table 3. Results of Equation 2 Regression with DC as the Base County

Regression Variable	Coef.	Robust Std.	t	P>t	[95% Conf. Interval]	
County						
AC	410.9	956.9	0.43	0.668	-1466	2287
FC	38.34	745.4	0.05	0.959	-1423	1500
MC	-611.9	1163	-0.53	0.599	-2892	1668
PG	-40.10	817.7	-0.05	0.961	-1644	1564
Tier	886.4	636.0	1.39	0.164	-360.9	2134
Size	**2.786**	**0.0640**	**43.57**	**<0.000**	**2.661**	**2.912**
County#Tier						
AC#Premium	1146	1810	0.63	0.527	-2403	4696
FC#Premium	511.0	1090	0.47	0.639	-1626	2648
MC#Premium	285.9	1325	0.22	0.829	-2313	2885
PG#Premium	-280.8	1104	-0.25	0.799	-2445	1884
County#Size						
AC	-0.167	0.117	-1.43	0.154	-0.396	0.0625
FC	-0.135	0.0856	-1.58	0.115	-0.303	0.0330
MC	-0.0865	0.131	-0.66	0.51	-0.344	0.171
PG	-0.129	0.0928	-1.39	0.164	-0.311	0.0528
Tier#Size						
Premium	0.122	0.0828	1.47	0.141	-0.0405	0.284
County#Tier#Size						
AC#Premium	-0.0325	0.188	-0.17	0.862	-0.401	0.336
FC#Premium	0.0310	0.120	0.26	0.796	-0.204	0.265
MC#Premium	0.0156	0.150	0.10	0.917	-0.279	0.310
PG#Premium	0.0682	0.122	0.56	0.577	-0.172	0.308
Constant	**1619**	**484.2**	**3.34**	**0.001**	**669.0**	**2568**

The results in Table 3 indicate that County nor Tier have a significant impact on the regression constant. When County is interacted with Tier, no interaction has a statistical impact. County-Size interactions appear to have no significant impact as well, possibly indicating no regional differences. The only variables that have any impact are the Size variable and the regression constant. By iterating through the base county, all significant relationships between counties can be determined12. This is unnecessary for the Tier variable since it only has two levels, so any hypothesis testing will only invert the sign of the test statistic, and under the normally distributed errors assumption will result in the same p-value. The same findings are apparent in every iteration.

While Table 3 is useful in looking at all possible combinations of the County, Tier, and Size variables, there is the possibility that it is overparametrized, confounding truly significant variables with non-significant variables. To test this concern, two less parametrized

[12] Regression Tables in Appendix A

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